

### 3. Fluid dynamics

#### 3.1. Fluid dynamics in astrophysics

Fluid dynamics is one of the main areas of study in astrophysics. It describes phenomena such as accretion disks, star formation, galactic dynamics or the interstellar medium.

Because of the computational and mathematical impossibility of studying the behaviour of the ensemble of particles that constitute fluids, fluid dynamics eases the study of such systems by studying the macroscopic behaviour of the fluid such that meaningful statistics can be recovered. Thus, fluids are defined by continuous fields of density, velocity and pressure, which are defined everywhere. This approximation works under the assumption that the mean free path of the particles that constitute the fluid is much smaller than the characteristic scale,  $\lambda_{mfp} \ll L$ . If this is not the case, a theory of kinematics is needed, given by Boltzmann's formula. In the following sections, the basic equations of hydrodynamics in the case of an inviscid fluid and no thermal conductivity are obtained and justified within the context of the interstellar medium.

##### 3.1.1. The Euler equations

Let's first consider an arbitrary closed volume  $V$  with a defined surface  $S$  which contains a fluid of mass  $M$  with a velocity  $\mathbf{v}$  defined at each point of the volume. By considering the amount of mass trapped by the volume  $V$ ,  $M$ , and the mass flowing out of this volume, which is defined by  $\rho \mathbf{v} \cdot d\mathbf{S}$  and relating them through the divergence theorem, we can write the continuity equation 26:

$$\partial_t \rho + \nabla \cdot (\mathbf{v} \rho) = 0. \quad (26)$$

This equation is interpreted as the conservation of mass within a certain volume. The change in the density depends on the amount of mass entering and leaving the volume.

Assuming a Newtonian and isotropic fluid, the total force applied to the fluid volume,

defined by the closed surface  $S$ , is given by the integral

$$F_{total} = - \oint_S p \, d\mathbf{s}, \quad (27)$$

of the pressure taken over the whole surface. By transforming the surface integral to a volume integral via the divergence theorem, we obtain that the force exerted on a volume element of fluid  $d\mathbf{V}$  is given by  $-\nabla p$ . Forward writing Newton's 2nd law on a unit volume of fluid, we get

$$\rho \frac{D\mathbf{v}}{Dt} = -\nabla p, \quad (28)$$

having introduced notation for Lagrangian or material derivatives  $\frac{D}{Dt}$ , this derivative is not evaluated at a fixed point in space but rather follows a parcel of fluid. The derivative at a fixed point in space is referred to as the Eulerian derivative  $\partial_t$  and is related to the Lagrangian derivative through  $\frac{D}{Dt} = \partial_t + \mathbf{v} \cdot \nabla$ . Since both are related by a simple equation, the entirety of hydrodynamics can be written in either of the frameworks. In the present work, we employ grid-based methods where, rather than tracking parcels, the parcel's location is fixed. Therefore, the Eulerian view is more convenient. Thus, we rewrite equation 28 in the Eulerian derivative framework and obtain

$$\rho \frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\nabla p. \quad (29)$$

The equations for the conservation of mass 26 and momentum 29 are referred to as Euler's equations, both written in the Eulerian derivative fashion. Together with the equation of state, they define the evolution of a fluid with zero viscosity and no thermal conductivity. These equations were derived with no external source terms assumed, like gravity in the case of the momentum equation. These terms can be added to the RHS of each equation to incorporate extra physics into our model.

Our system so far lacks closure, to do so, we add an equation of state to it. A typical choice of equation of state is the ideal gas equation of state

$$P = (\gamma - 1) \rho \mathcal{U}, \quad (30)$$

where  $\mathcal{U}$  is the internal energy per unit mass and  $\gamma$  is the adiabatic index given by  $5/3$  in the case of monatomic gases. As a new field was introduced, the internal energy, we must add another equation to complete the system. The conservation of energy equation

$$\frac{\partial \rho e}{\partial t} + \nabla \cdot [(\rho e + P)\mathbf{v}] = 0, \quad (31)$$

which can be derived from the same arguments as equations 26 and 29, given that now the conserved quantity is the total energy per unit mass  $e$  given by  $e = \mathcal{U} + \frac{1}{2}\mathbf{v}^2$ . Lastly, with this equation we've given closure to the hydrodynamical model. The list of assumptions considered while obtaining these equations and their validity in modelling the astrophysical interstellar medium is as follows <sup>5</sup>:

- **Hydrodynamic approximation:** Given the Hydrogen number density  $n_H \sim 0.6 \text{ cm}^{-3}$  the mean free path is  $\lambda_{mfp} \sim 2 \times 10^{15} \text{ cm}$ . Whereas the characteristic length  $L \sim 1 \text{ pc} = 3.09 \times 10^{18} \text{ cm}$  is 3 orders of magnitude larger than  $\lambda_{mfp}$ . Thus justifying the hydrodynamic approximation.
- **Inviscid:** The inviscid assumption can be validated by comparing the viscous diffusion and dynamic timescales. Following [Carraro, 2021][Chapter 4.1], for WNM conditions the viscous diffusion timescale is  $\tau_\nu \sim 2.5 \times 10^7 \text{ yr}$  whereas the dynamical timescale is  $\tau_{dyn} \sim 1.2 \times 10^5 \text{ yr}$ . Then  $\frac{\tau_\nu}{\tau_{dyn}} \sim 200 \gg 1$  meaning that viscous momentum transport occurs much more slowly than the dynamical evolution and can therefore be neglected.
- **Negligible thermal conductivity:** Similarly, the thermal conductivity assumption can be validated by comparing characteristic times. Again, following [Carraro, 2021], for WNM values the dynamic thermal timescale  $\tau_t \sim 1.4 \times 10^8 \text{ yr}$ , thus  $\frac{\tau_t}{\tau_{dyn}} \sim 10^3 \gg 1$ . In consequence, the thermal conductivity can be neglected as the dynamical evolution of the gas dominates.
- **Ideal gas:** The ISM is a highly diffuse gas. The hydrogen number densities  $n_H \sim$

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<sup>5</sup>We take the values from the warm neutral medium WNM from [Draine, 2011][Table 1.3] with  $n_H \simeq 0.6 \text{ cm}^{-3}$  and  $T \simeq 5000 \text{ K}$ . This phase constitutes around 40% of the ISM volume, thus being representative of the fluid dynamics considered.

$10^{-3} - 10^6 \text{ cm}^{-3}$ , thus collisional interactions between particles are scarce, and the ideal gas assumption is well motivated.

**A comment on simulations** Equations for the mass 26, momentum 29 and energy 31 conservation together with the energy and the equation of state give closure to the model. To simulate such a fluid, the conserved quantities are evolved over time by defining a system of conserved equations

$$\partial_t S + \nabla \cdot F(S) = 0, \quad (32)$$

where  $S$  is the conserved state vector and  $F$  the flux rank 2 tensor. With the equations described prior,  $S$  and  $F$  are

$$S = \begin{pmatrix} \rho \\ \rho \mathbf{v} \\ \rho e \end{pmatrix}, \quad F(S) = \begin{pmatrix} \rho \mathbf{v} \\ \rho \mathbf{v} \mathbf{v}^\top + P \mathbf{I} \\ (\rho e + P) \mathbf{v} \end{pmatrix} \quad (33)$$

where  $\mathbf{I}$  is the identity tensor. To write these equations, the conservative form of the momentum equation was used. For a derivation of such an equation, see [Sormani, 2022, Section 1.8]. The first component corresponds to the mass flux, the second to the momentum flux and the last one to the energy flux. Written in this conservative form, the Euler equations become a system of hyperbolic conservation laws, a formulation well suited for finite volume methods commonly employed in computational fluid dynamics.

### 3.1.2. Magneto hydrodynamics

As the interstellar medium is weakly ionised, we must consider the electromagnetic fields that are coupled to the hydrodynamics equations. The theory that describes such an ionised fluid is magnetohydrodynamics (MHD). In the case of the ISM, the ionisation fraction  $x_e = n_e/n_H$  varies across the considered areas. In dense molecular clouds, the gas is almost neutral,  $x_e \lesssim 10^{-6}$ , whereas in HI or HII regions the ISM is partially to

fully ionised,  $x_e \sim 10^{-3} - 1$  [Draine, 2011, Chapter 13].

Maxwell's equations that model the EM fields are

$$\nabla \cdot \mathbf{E} = 4\pi\rho_e \quad (34)$$

$$\nabla \cdot \mathbf{B} = 0 \quad (35)$$

$$\nabla \times \mathbf{E} = -\frac{1}{c} \partial_t \mathbf{B} \quad (36)$$

$$\nabla \times \mathbf{B} = \frac{4\pi}{c} \mathbf{J} + \frac{1}{c} \partial_t \mathbf{E}, \quad (37)$$

first assuming the fluid considered has a net charge of zero and therefore there's almost no net force exerted by  $\mathbf{E}$ , we can discard the time derivative of the electric field and write  $\nabla \times \mathbf{B} = \frac{4\pi}{c} \mathbf{J}$ . Thus, writing the Lorentz force  $F_{EM} = \frac{1}{c} \mathbf{J} \times \mathbf{B}$ <sup>6</sup>. Taking the induction equation and substituting  $\mathbf{J}$ , the Lorentz force can be written in a conservative form and only in terms of the magnetic field

$$F_{EM} = \frac{1}{4\pi} (\nabla \times \mathbf{B}) \times \mathbf{B} = \frac{1}{4\pi} \left[ -\frac{1}{2} \nabla B^2 + (\mathbf{B} \cdot \nabla) \mathbf{B} \right] = \frac{1}{4\pi} \nabla \left[ -\frac{1}{2} B^2 + \mathbf{B} \mathbf{B}^\top \right], \quad (38)$$

where on the first equality the following vector identity was used  $(\nabla \times \mathbf{B}) \times \mathbf{B} = -\frac{1}{2} \nabla B^2 + (\mathbf{B} \cdot \nabla) \mathbf{B}$  and on the second one  $\nabla \cdot (\mathbf{B} \mathbf{B}^\top) = (\mathbf{B} \cdot \nabla) \mathbf{B} + \mathbf{B} (\nabla \cdot \mathbf{B}) = (\mathbf{B} \cdot \nabla) \mathbf{B}$  by the Gauss magnetic field equation. We can insert the Lorentz force equation in the momentum equation 29 and obtain:

$$\rho \left( \frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} \right) = -\nabla \left( P + \frac{B^2}{8\pi} \right) + \frac{1}{4\pi} (\mathbf{B} \cdot \nabla) \mathbf{B}. \quad (39)$$

Using the last form of 38 and the momentum part from 33, the conservative form of the ideal MHD momentum equation can be written

$$\frac{\partial \rho \mathbf{v}}{\partial t} + \nabla \left[ \rho \mathbf{v} \mathbf{v}^\top + \left( P + \frac{B^2}{8\pi} \right) \mathbf{I} - \frac{1}{4\pi} \mathbf{B} \mathbf{B}^\top \right] = 0. \quad (40)$$

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<sup>6</sup>This result is also obtained by assuming an almost net charge of zero and no relativistic effects as done in [Sormani, 2022]. The third equation in 34 shows that  $E \sim \frac{v}{c} B$  and therefore, when compared with the LHS of the fourth equation, the term  $\partial_t \mathbf{E}$  can be neglected. Depending on the author one assumption is mentioned over the other, whichever is chosen is important to notice that by neglecting  $\partial_t \mathbf{E}$  Maxwell's equations and the Euler equations are both Galilean invariant.

In order to ensure the closure of the model, the induction equation under the assumption of a perfect conductor ( $\sigma \rightarrow \infty$ )

$$\frac{\partial \mathbf{B}}{\partial t} = \nabla \times (\mathbf{v} \times \mathbf{B}) \quad (41)$$

is needed. Using this equation, by studying the flux that goes through a given surface, it can be shown, that the field lines are "frozen" in the fluid, as shown in [Draine, 2011, Chapter 35.4]. We take from the same reference the motivation to set the assumption of a perfect conductor in the case of an ionised medium. The authors find the time scale for magnetic field decay on an ionised gas with sizes  $L \gtrsim 10^2 \text{AU}$  to be larger than the age of the Universe, hence motivating the assumption for large structures, whereas on short scales this approximation no longer applies.

Lastly, we must consider the conservation of energy equation, considering the magnetic energy

$$\partial_t \left( \rho e + \frac{B^2}{8\pi} \right) + \nabla \cdot \left[ \left( \rho e + P + \frac{B^2}{8\pi} \right) \mathbf{v} - \frac{1}{4\pi} (\mathbf{v} \times \mathbf{B}) \times \mathbf{B} \right] = 0, \quad (42)$$

where  $e$  is given by the internal and kinetic energy. For a derivation of this equation, see [Galtier, 2016, Section 3.3].

All the equations considered in these last two subsections are aggregated in Table 1.

Table 1: Hydrodynamic and ideal MHD equations in conservative form. Here  $\rho$  is the density,  $\mathbf{v}$  the velocity field,  $P$  the thermal pressure,  $\mathbf{B}$  the magnetic field, and  $e = \mathcal{U} + \frac{1}{2}v^2$  the total hydrodynamic energy per unit mass.

| Equation      | Hydrodynamics (HD)                                                                                | Ideal MHD                                                                                                                                                                                                         |
|---------------|---------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Continuity    | $\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0$                                            | $\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0$                                                                                                                                                            |
| Momentum      | $\partial_t(\rho \mathbf{v}) + \nabla \cdot (\rho \mathbf{v} \mathbf{v}^\top + P \mathbf{I}) = 0$ | $\partial_t(\rho \mathbf{v}) + \nabla \cdot \left[ \rho \mathbf{v} \mathbf{v}^\top + \left( P + \frac{B^2}{8\pi} \right) \mathbf{I} - \frac{\mathbf{B} \mathbf{B}^\top}{4\pi} \right] = 0$                        |
| Energy        | $\partial_t(\rho e) + \nabla \cdot [(\rho e + P) \mathbf{v}] = 0$                                 | $\partial_t \left( \rho e + \frac{B^2}{8\pi} \right) + \nabla \cdot \left[ \left( \rho e + P + \frac{B^2}{8\pi} \right) \mathbf{v} - \frac{1}{4\pi} (\mathbf{v} \times \mathbf{B}) \times \mathbf{B} \right] = 0$ |
| Eqn. of state | $P = (\gamma - 1) \rho \mathcal{U}$                                                               | $P = (\gamma - 1) \rho \mathcal{U}$                                                                                                                                                                               |
| Induction     | —                                                                                                 | $\partial_t \mathbf{B} = \nabla \times (\mathbf{v} \times \mathbf{B})$                                                                                                                                            |
| Constraint    | —                                                                                                 | $\nabla \cdot \mathbf{B} = 0$                                                                                                                                                                                     |

### 3.2. Differentiable MHD simulations

To simulate the Euler equations presented in Table 1 a number of proposed methods exist, these can be split into two groups, particle and grid based. Particle based methods track fluid particles (which are not the same as the physical particles that constitute matter) position and the macroscopic fluid quantities are obtained through kernel density estimation or any other field estimation method. On the other hand grid based methods divide the simulation box into a grid (either uniform or nonuniform). These cells can be either considered as a finite volume over which an average fluid field quantity value is assigned, known as finite volume methods, or, the centre of each cell can be considered a fixed point on which the fluid is evaluated, known as finite difference methods.

Magnetohydrodynamics simulations pose an extra challenge to the simulation, as a discretised form of the free magnetic field divergence has to be ensured throughout the simulation. If this is not the case, the simulation could lead to nonphysical results such as negative pressure or density, as proven in [Wu, 2018] Theorem 4.3. Thus, ensuring the divergence free (DF) condition is key to a physically stable simulation, a plethora of methods exist to ensure DF such as constrained transport, projection methods or divergence cleaning. For a more detailed explanation of each of the methods, we refer the reader to the introduction of [Hopkins, 2016].

#### 3.2.1. Differentiable simulations

End-to-end differentiable simulations have appeared recently as a new paradigm of simulations leveraged by the advancements in automatic differentiation codes such as JAX or PyTorch. Consider an initial discretised state determined by a set of parameters  $\theta$ ,  $s_0(\theta)$  of a dynamical system.  $s_1(\theta)$  is then obtained by applying the discretized PDE operator  $\tilde{\mathcal{P}}$ ,  $s_1(\theta) = \tilde{\mathcal{P}}(s_0(\theta))$ . By iteratively applying  $\tilde{\mathcal{P}}$ , the final state  $s_n$  can be obtained

$$s_n(\theta) = \tilde{\mathcal{P}} \circ \tilde{\mathcal{P}} \dots \circ \tilde{\mathcal{P}}(s_0(\theta)) = \tilde{\mathcal{P}}(\tilde{\mathcal{P}}(\dots \tilde{\mathcal{P}}(s_0(\theta))\dots)) = \tilde{\mathcal{P}}^n(s_0(\theta)) \quad (43)$$

where the  $n$  exponential signifies repeated application. Assuming a differentiable simulation and given a scalar loss function,  $\mathcal{L}(s_n, s_n^*)$  where  $s_n^*$  is a target final state, we can



compute  $\partial\mathcal{L}/\partial\theta$  and apply any gradient based optimisation method to optimise for  $\theta^*$ . Examples of applications of these methods are:

- Inverse problems: Given an initial state parametrized by a set of parameters  $\theta$  and a desired final state we can optimize the parameters such that the initial state evolves to the target final state, such as in [Yang et al., 2025] where a differentiable simulation was used together with a robot arm to infer physical parameters of materials.
- Trajectory optimisation: Typically used in simulations that consist of objects and entities with parameters of their own (e.g. simulations of robots), where the path of such entities is optimised by reducing a cost function. Such as in [Du et al., 2021] where robots were taught to walk on their four legs by minimising a cost function through the differentiable simulation.
- Neural network augmented simulations: Hybrid AI-simulations can be obtained with differentiable simulations where a neural network is embedded with the simulation as a means of improving either performance, accuracy or both, an example of such a neural network integration was used in [Um et al., 2021] where by intertwining steps of the simulation and a corrector neural network more accurate physical statistics could be recovered from low resolution simulations.

For a further review of applications of differentiable simulations, we refer the reader to the [Newbury et al., 2024] review. Differentiable simulations examples can be found in material sciences ([Dryschev et al., 2025], [Daviet et al., 2025], [Daubner et al., 2025], [Mahlau et al., 2026]), quantum and particle sciences ([Kaiser et al., 2024], [Guilmin et al., 2025]), astronomy and cosmology ([Horowitz et al., 2025], [Vermariën et al., 2025], [List et al., 2025], [Viterbo and Buck, 2025]) or biophysics [Neymotin and Hazan, 2025] to mention some examples.

### 3.2.2. Astronomix: differentiable MHD interstellar simulation written in JAX

The simulator used in this work is the MHD simulation code Astronomix [?] (formerly known as jfluids [Storeks and Buck, 2024]). Astronomix implements finite volume sim-

ulations with Lax-Friedrichs, HLL and HLLC Riemann solvers and recently finite difference simulations. Astronomix has a focus on astronomical applications and implements physical modules such as self-gravity, cosmic rays or stellar winds. Completely written in JAX, Astronomix allows for differentiable simulation applications like the previously discussed ones.

JAX [Bradbury et al., 2018] is a Python library which allows for computations with arrays (with NumPy [Harris et al., 2020] like syntax) and allows for Just-In-Time (JIT) compilation. JIT allows for Open XLA compilation of any pure function written in JAX (pure signifies the lack of side effects like print statements), such compilation allows for high speed functions that can run on CPUs, GPUs or TPUs. JAX also allows for automatic differentiation through any pure function, which allows for high performance differentiable simulation pipelines.

### 3.2.3. Solver-in-the-loop

One common machine learning application on simulations within the literature consists of training a network to emulate the simulation, thus skipping the whole PDE applications, like in [Hao Wu, 2026] where an operator model is trained as a surrogate for turbulence simulations. Such an application, although proven useful, has its shortcomings, especially under long simulations where the errors will grow, making the emulated simulation unphysical. The solver-in-the-loop approach, proposed by Um et al. in [Um et al., 2021], instead, doesn't substitute the simulator but adds an extra term in between applications of the PDE. The model interacts only with the local structures not captured by the simulator and thus acts as a residual term. Consequently, the learning objective is reduced to a drift term in the simulation space towards the target state, as shown in the schematic 3.1. This target state is established at training, and during inference we expect the model to have learned the discrepancies between the simulated and target dynamics.

A simulation can typically be described as an iterative application of the discretised PDE operator  $\mathcal{P}$  to an initial state  $s_{t=0}$ . Given an initial state with resolution HR  $s_{t=0|HR}$ , if we

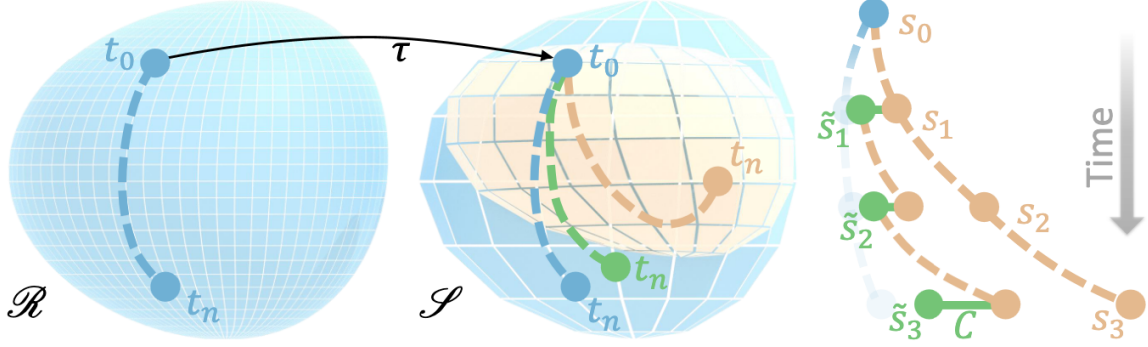


Figure 3.1: Solver-in-the-loop phase space diagram. The left sphere represents the HR phase space; by downscaling the initial state with transformation  $\tau$ , the state is moved to the LR phase space on the right. Evolving this state results in the low resolution trajectory in orange. Correcting now in between applications of the PDE with corrector  $\mathcal{C}$  the green trajectory is obtained. Figure taken from [Um et al., 2021].

want to get the state at time  $T$  we can apply the PDE  $n$  times <sup>7</sup>  $s_{t=T|HR} = \mathcal{P}^n(s_{t=0|HR})$ , where the exponent signifies iterative application. If now through a nonaffine operation  $\tau$  we reduce the resolution of the state to LR  $\tau(s_{t=0|HR}) = s_{t=0|LR}$  and apply the same time evolution to this state  $s_{t=T|LR} = \mathcal{P}^n(s_{t=0|LR})$  we will find that  $\tau(s_{t=T|HR}) \neq s_{t=T|LR}$ . This is typically expected, and  $\tau(s_{t=T|HR})$  might not even be initially accessible in phase space for the low resolution simulation, even more so in chaotic systems the low resolution simulation would be further from the high resolution resolved one in phase space. These can all be attributed to the numerical errors introduced when discretising, and as such it is assumed that the higher the resolution, the closer we are to the actual physical solution of the system.

We can then introduce a corrector term in between applications of the PDE,  $\mathcal{C}$ , such that the iterative simulation is  $(\mathcal{P}\mathcal{C})^n s_{t=0} = \tilde{s}_{t=T}$ . The corrector, a neural network, learns the inherent numerical error introduced by the discretisation and changes the trajectory of the simulation in phase space towards the highly resolved downsampled states, as shown in Figure 3.1 green line. Thus  $\mathcal{C}$  is tasked with learning a drifting term in the state space only through residual terms. In this way, the model has less responsibility, and

<sup>7</sup>Here we are not going to make the distinction between fixed or variable timestep

we can exploit its full expressive power in achieving better accuracy for a fraction of the performance. Lastly it's worth commenting on how these models are used and trained, when treating the PDE simulation as an output system (from the point of view of the corrector), the training procedure resembles the one from a Recurrent Neural Network [Schmidt, 2019], where the model iteratively gets an input which in our case gets modified afterwards by the discretised PDE operator  $\mathcal{P}$ , thus known training strategies and overall knowledge from RNNs can be exploited to train solver-in-the-loop (SOL) models.

### 3.3. MHD type problems

In the following we introduce 3 MHD type problems used to test simulation capabilities: complex magnetic fields structures, wave shocks, and chaotic systems.

#### 3.3.1. Blasts

Shock waves occur in the ISM through a lot of different mechanisms (e.g. supernova explosions, expanding HII regions, accelerated gases due to gravity). Shock waves are disturbances that propagate faster than the sound speed of the medium and originate due to a discontinuity. The conditions under which a discontinuity in the medium turns into a shock wave are given by the Rankine–Hugoniot conditions [Galtier, 2016, Section 6.1].

We are concerned with MHD Blasts as they provide challenging MHD simulations due to the high Reynolds number and plasma  $\beta$ . Blast waves typically occur in supernova explosions both in the initial collapse and afterwards the revived shock. Thus, it's important to understand this problem as it determines the outward flow of mass and energy from supernova explosions.

We take the initial conditions for a MHD Blast from [Mignone and Del Zanna, 2021]. The state is set on a  $[0, 1]^3$  box, the initial density and velocity are constant with  $\rho_0 = 1.0$  and  $\mathbf{v}_0 = 0$ , the pressure is constant everywhere  $p_0 = 1$  except for a spherical region in the center of the simulation box  $p_{center} = 100.0$  with radius  $r_0 = 0.125$ . The magnetic field is given by  $\mathbf{B} = B_0 (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$  with  $B_0 = 10.0$  and  $\theta = \pi/2, \phi =$

$\pi/2$ , therefore the ambient plasma  $\beta = 0.02$ , resulting in a highly magnetic dominated environment. Depending on the implementation a soft gradient from the high density sphere to the low density environment is also implemented.

$$p(r) = \begin{cases} p_0 + (p_{\text{center}} - p_0) \frac{r_1 - r}{r_1 - r_0}, & r_0 < r < r_1 \\ p, & r \leq r_0 \text{ or } r \geq r_1 \end{cases} \quad (44)$$

where  $p$  is the initialised pressure before implementing the gradient and  $r_1 = 0.1375$ . We evolve the system till  $t = 0.2$ . An example of an MHD blast setup described here is given in Figure 3.2.

The chosen parameters don't fully represent a supernova setup, rather a numerical setup by emphasizing the magnetic effects. Although the setup doesn't represent an accurate supernova progenitor, we capture the basic structure of an expanding supernova blast, a localised high pressure region in a magnetised environment. The resulting shock interacts with the magnetic field to test the magnetohydrodynamic capabilities of the code, as the ambient is highly magnetised compared with a typical supernova ambient.

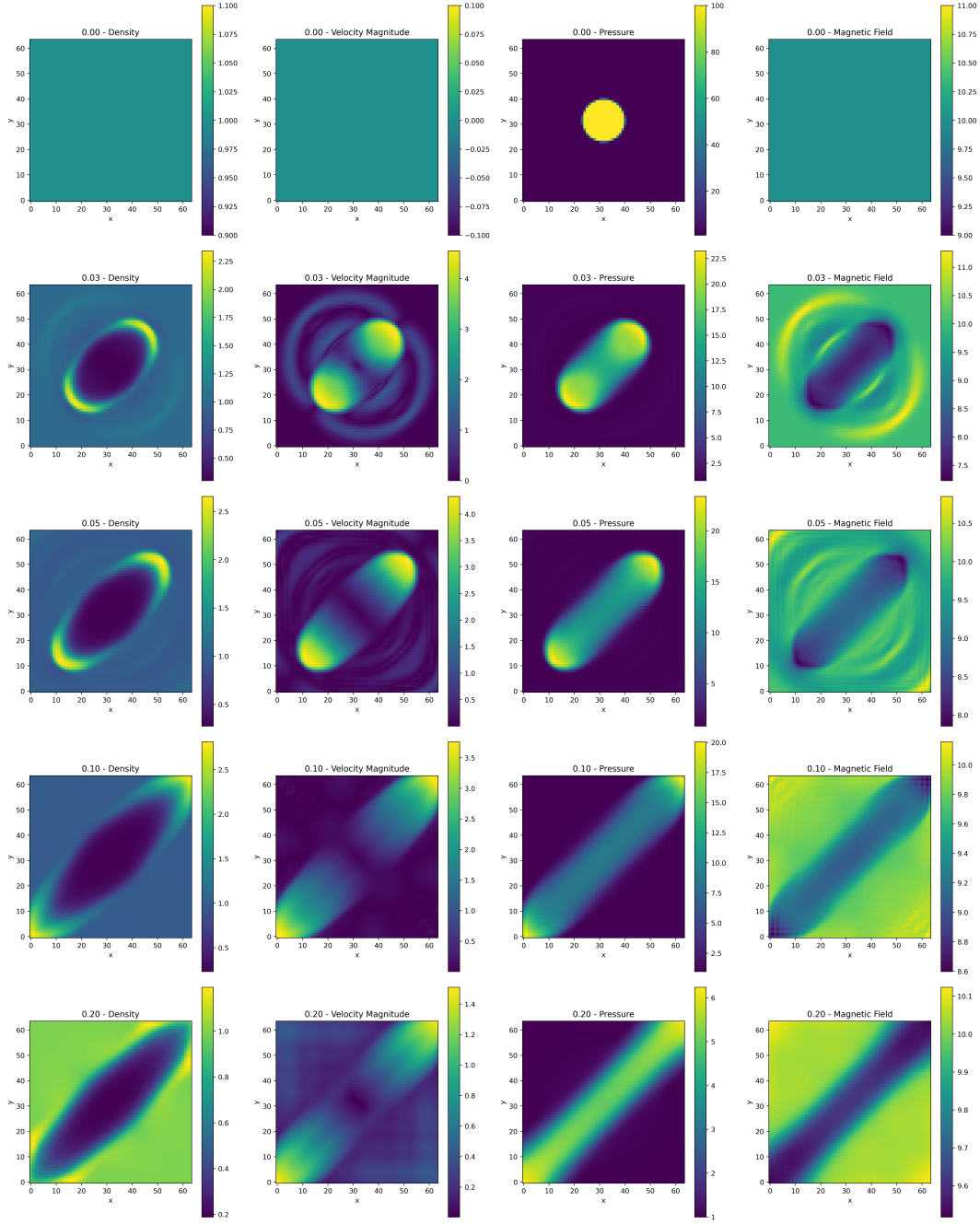


Figure 3.2: 3D MHD Blast solved with Astronomix. Resolution is  $64^3$ , solver mode is finite volume, and boundary conditions are open. The times shown are  $[0.00, 0.03, 0.05, 0.10, 0.2]$ , with each time shown in a separate row, to clearly see the initial shock waves. Note that the color scalings are different on each row. The rest of the setup details are given in section 3.3.1.

### 3.3.2. Orszag-Tang vortex

A key problem in testing the transition to supersonic MHD turbulence is the Orszag-Tang vortex introduced in [Orszag and Tang, 1979]. It also tests the DF condition, MHD shocks and shock-shock interaction [Lopes et al., 2018]. The original problem was written for 2 dimensions, here we expand it to 3, following [Helzel et al., 2011]. The initial conditions are given by a constant density and pressure,  $\rho_0 = \gamma^2$ ,  $p_0 = \gamma$ , where  $\gamma$  is the adiabatic index, a variable magnetic field  $\mathbf{B} = (-\sin(y), -\sin(2x), 0)$ , and a velocity field given by 2. The computational domain is  $[0, 2\pi]$  in each axis of the box, with a final time of  $t = \pi$ . An image comparing the initial and final state of the OT vortex can be seen at 3.3.

Table 2: Orszag-Tang vortex: initial condition for velocity field.

|       | 2D         | 3D                                     |
|-------|------------|----------------------------------------|
| $u_x$ | $-\sin(y)$ | $-[1 + \varepsilon_p \sin(z)] \sin(y)$ |
| $u_y$ | $\sin(x)$  | $[1 + \varepsilon_p \sin(z)] \sin(x)$  |
| $u_z$ | 0          | $\varepsilon_p \sin(z)$                |

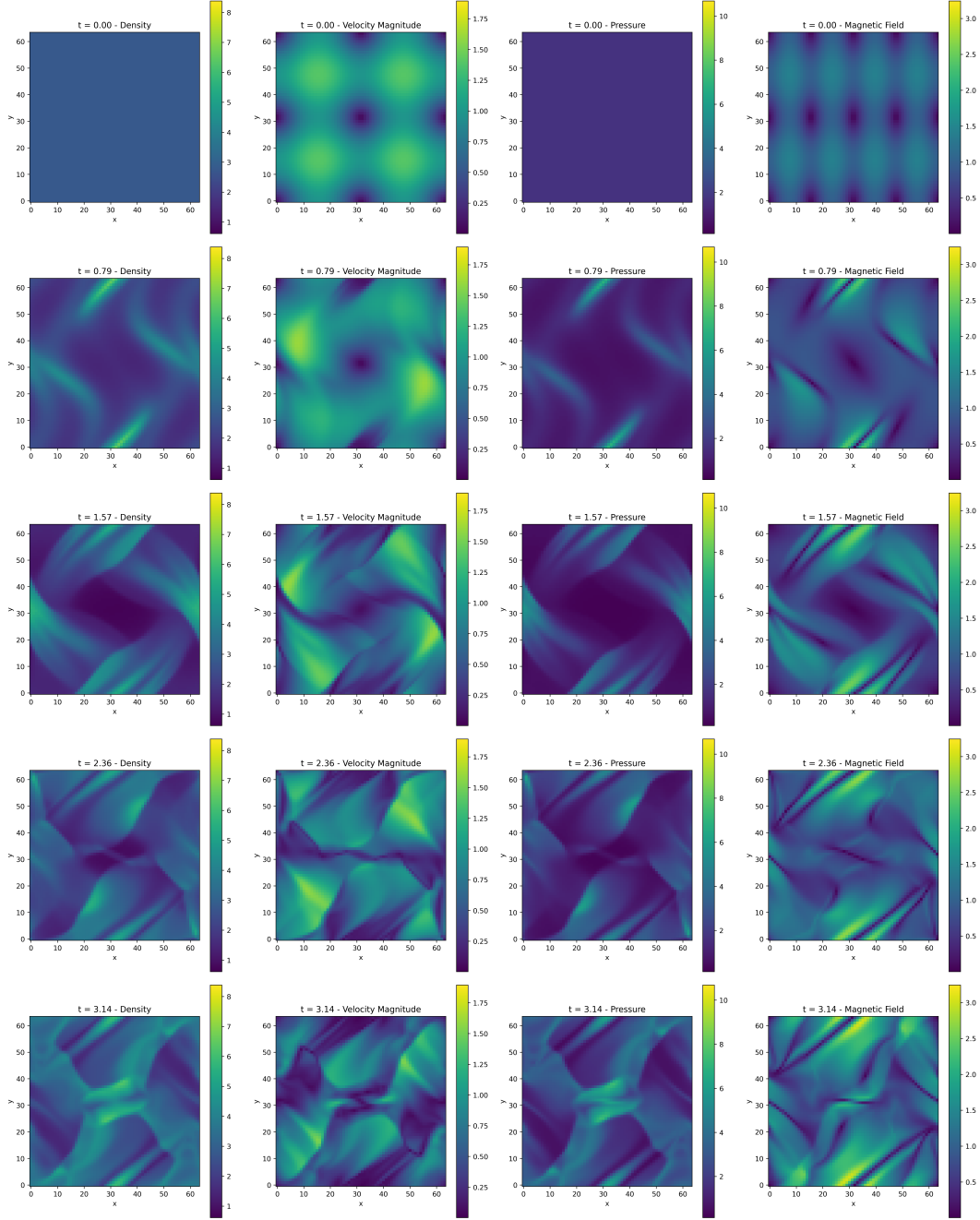


Figure 3.3: Orszag-Tang vortex state evolution at  $t = [0, \pi/4, \pi/2, 3\pi/4, \pi]$ , with each time shown in a separate row, computed in Astronomix with  $\gamma = 5/3$ . The simulation was computed in finite difference mode with periodic boundaries and a resolution of  $64^3$ .



### 3.3.3. Turbulence

Flow regimes are broadly categorised into two: laminar and turbulent. Considering the Navier-Stokes equations, where the Euler equations are the inviscid approximation of them, turbulence appears when the inertial nonlinear effects of the Navier-Stokes equations dominate over the viscosity effects. The kinematic viscosity of the fluid  $\nu$  appears in the equation as a term  $\nu \nabla^2 v$ . The competition between this term and the nonlinear velocity term  $v \nabla v$  is what creates turbulence. This competition is parametrised by the Reynolds number  $Re = uL/\nu$ , where  $u$  is the flow velocity, and  $L$  is the characteristic length. For a given geometry and viscosity, low enough Reynolds number values correspond to a laminar regime, whereas a high enough value corresponds to a turbulent regime<sup>8</sup>.

Turbulent flows contain a hierarchy of eddies spanning a wide range of scales, from large energy-containing motions to small dissipative structures. The large scales usually contain the most kinetic energy and are anisotropic, a consequence of the external forces that caused the turbulence, and define the characteristic length of the fluid  $L$ . The smaller scales are where the diffusion effects take place, this scales are characterized by the Kolmogorov scale  $\eta$  which denotes the smallest scales before the viscosity dominates and the kinetic energy is diffused. Hence there exists an energy cascade from the larger scales to the smallest ones, till the Kolmogorov scale is reached at which point diffusion takes effect and the kinetic energy is transformed into thermal.

Within the inertial range ( $\eta \ll r \ll L$ , where  $r$  is the separation scale), the diffusion effects are negligible, and therefore we can assume a mean energy cascade rate  $\epsilon$  through the range, which is equal to the diffusion rate and the supply energy rate. Kolmogorov's hypothesis states that velocity fluctuations in the inertial range depend only on  $r$  and  $\epsilon$ , resulting in  $\delta u(r) \propto (\epsilon r)$ . Hence, so does the energy spectrum  $E(k)$ ,

$$E(k) = C_K \epsilon^{2/3} k^{-5/3}, \quad (45)$$

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<sup>8</sup>The exact value under which the fluid transitions from one regime to the other highly depends on the geometry of the problem. In [Mukund and Hof, 2018] they give a value for this transition around 2000 for a pipe geometry.

where  $C_K$  is the Kolmogorov constant with a value of  $1.5 \sim 1.7$ . Therefore, recreating the correct energy spectra is important in numerical simulations of turbulence, as it represents whether the model is able to capture the correct rate of energy cascade across all scales.

In simulations governed by the Euler equations, an explicit physical viscosity term does not exist. Nevertheless, the discretisation of the PDEs introduces numerical diffusion, which acts as a subgrid scale model for energy diffusion at unresolved scales. Hence, we also observe turbulent regimes in these simulations, although the viscosity value is dependent on the simulation's resolution and scheme.

Two approaches exist for modelling subgrid scale diffusion in simulations. In implicit large eddy simulations (ILES) the effect is governed by the numerical diffusion, thus in this simulations the energy cascade and diffusion rate are both physically correct but the scale at which the diffusion happens to occur is wrong [Semenov, 2025]. The alternative is known as Explicit LES, where the sub grid scale diffusion is modelled through a closure term this can be done in an analytical form like eddy diffusivity models, mean dissipation formulations or dynamic procedures (for a more in depth analysis of analytical subgrid diffusion models we refer the reader to [Moser et al., 2021]) or modelled after data based models [Choi et al., 2025].

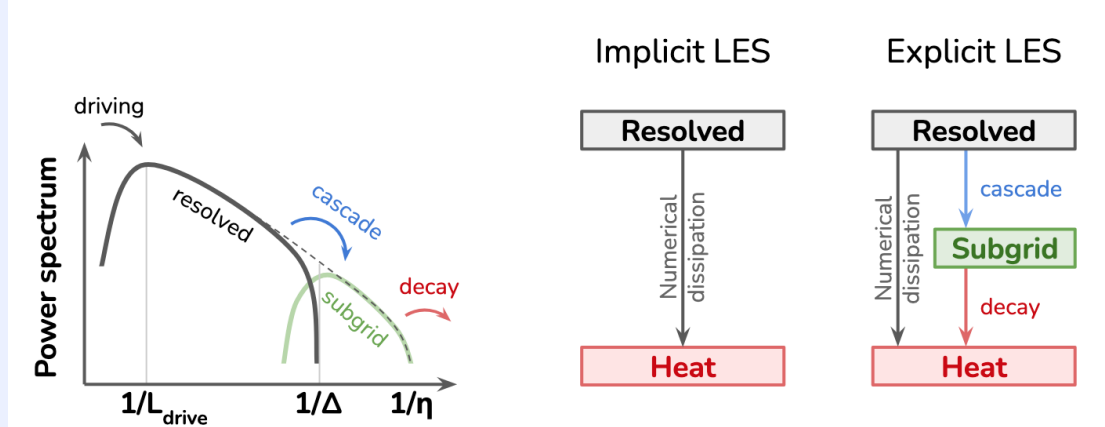


Figure 3.4: Scheme of the energy cascade and diffusion in Explicit and Implicit LES. Image taken from [Semenov, 2025].